



Oil Sands Process-Affected Water Toxicity Attribution and Evaluating Ageing as a Remediation Strategy

MSc Defense Presentation

Ian Gault* and Dr. Jonathan Martin

January 22nd, 2019



UNIVERSITY OF ALBERTA
FACULTY OF MEDICINE & DENTISTRY

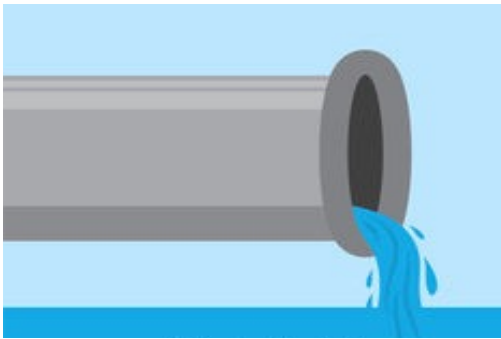
Department of Laboratory Medicine & Pathology

Base Mine Lake (BML), Fort McMurray, Alberta, 2015

Oil sands process-affected water (OSPW)¹

- A byproduct of the extraction process used to separate bitumen from sand, currently >1 billion m³

End Pit Lake Strategy²



Tailings Pond

Ageing



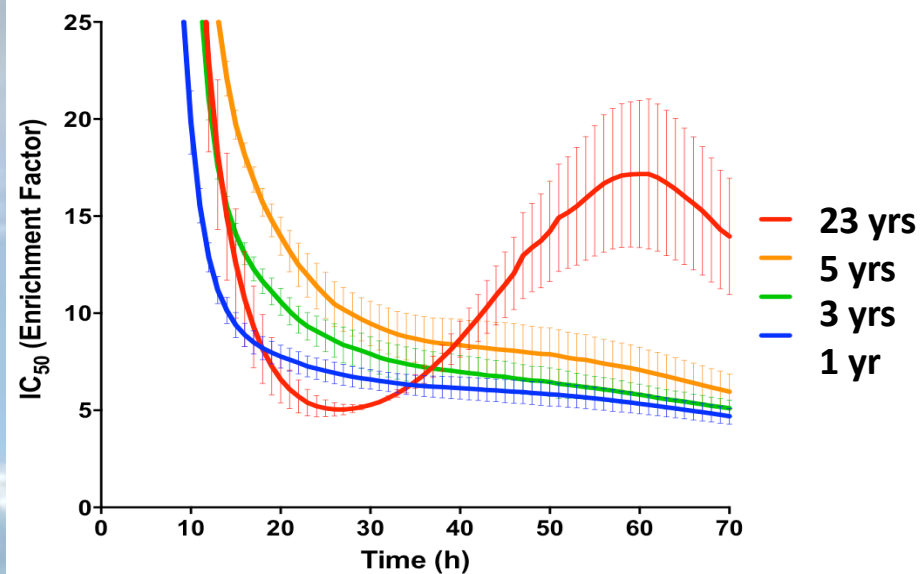
Hoped (but not proven): naturally detoxify via sedimentation and degradation; permanent feature connected to watershed³



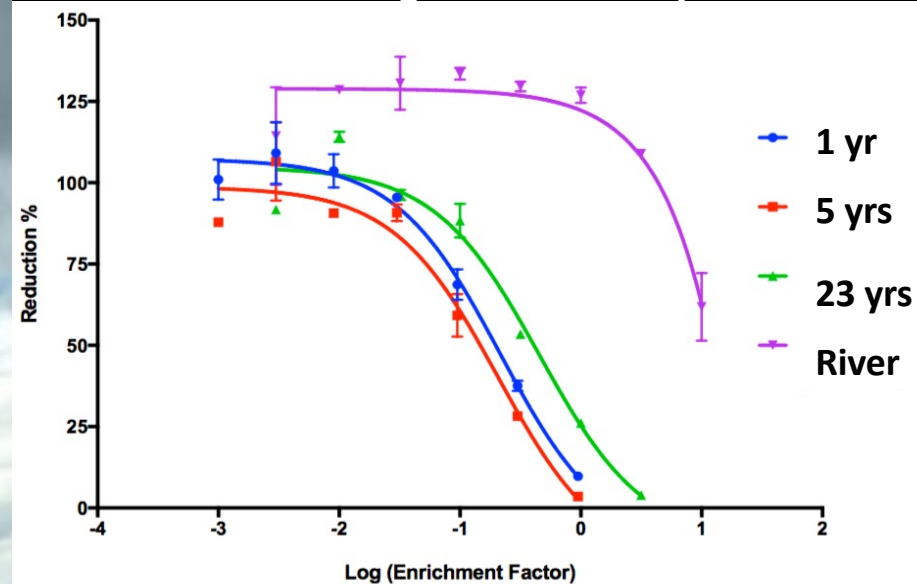
End Pit Lake

- **BML** commissioned in 2012 is the ONLY oil sand End Pit Lake

Real Time Cell Analysis



YAS Antagonism Example



	YES	YAS
Agonism	✗	✗
Antagonism	✓	✓

Years Aged: 1 yr

3 yrs

5 yrs

23 yrs

River
2017

Fresh

2013
BML

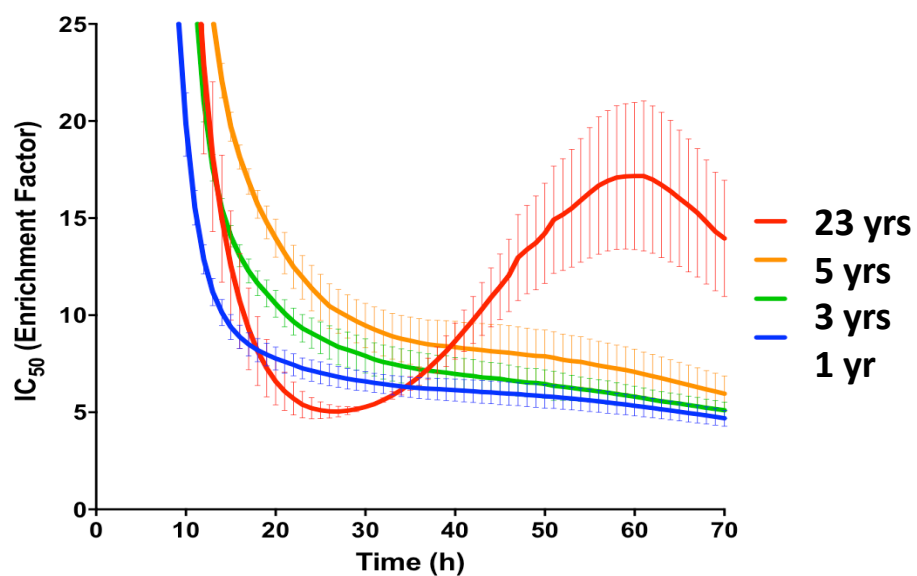
2015
BML

2017
BML

Pond
9

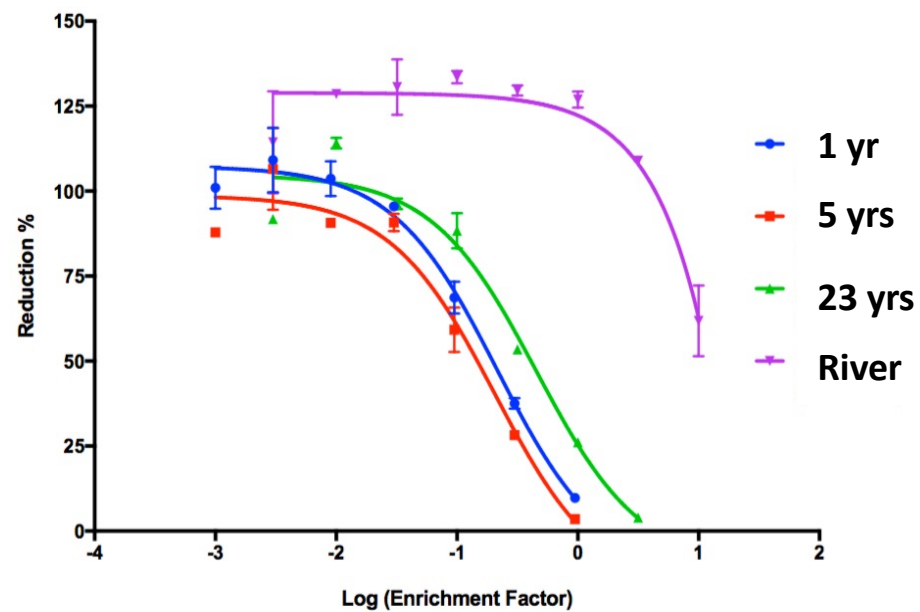
Aged

Real Time Cell Analysis



	YES	YAS
Agonism	✗	✗
Antagonism	✓	✓

YAS Antagonism Example





Before BML, preliminary *in situ* **Experimental Ponds** at Syncrude Canada tested strategy

Concerns on Effectiveness - Royal Society of Canada¹

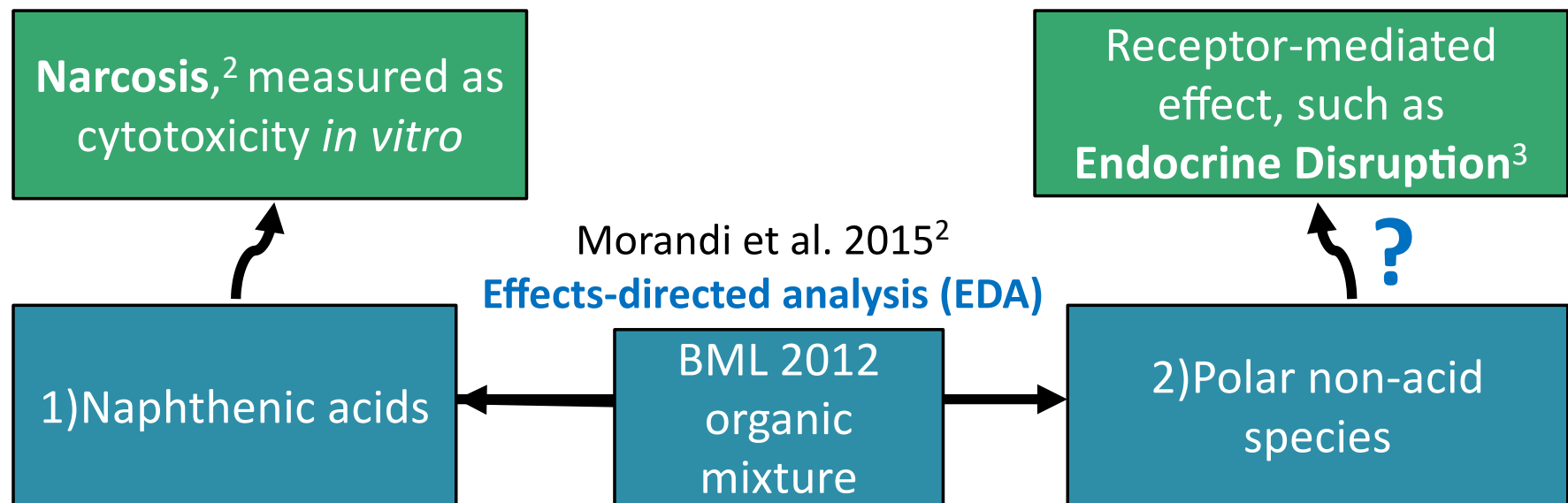
- Rate of degradation of organics is slow or unknown²
- Observations of reproductive and developmental effects in fish³ and midges⁴ exposed to aged OSPW

Release of BML is inevitable and necessary⁵

- Integration = unknown exposure and effects
- Downstream human communities concerned about own **health⁶**

Supercomplex Organic Mixture

- Largely uncharacterized,¹ and more easily described by variations in the formula $C_xH_yO_zS_aN_b$ (“Species”)²

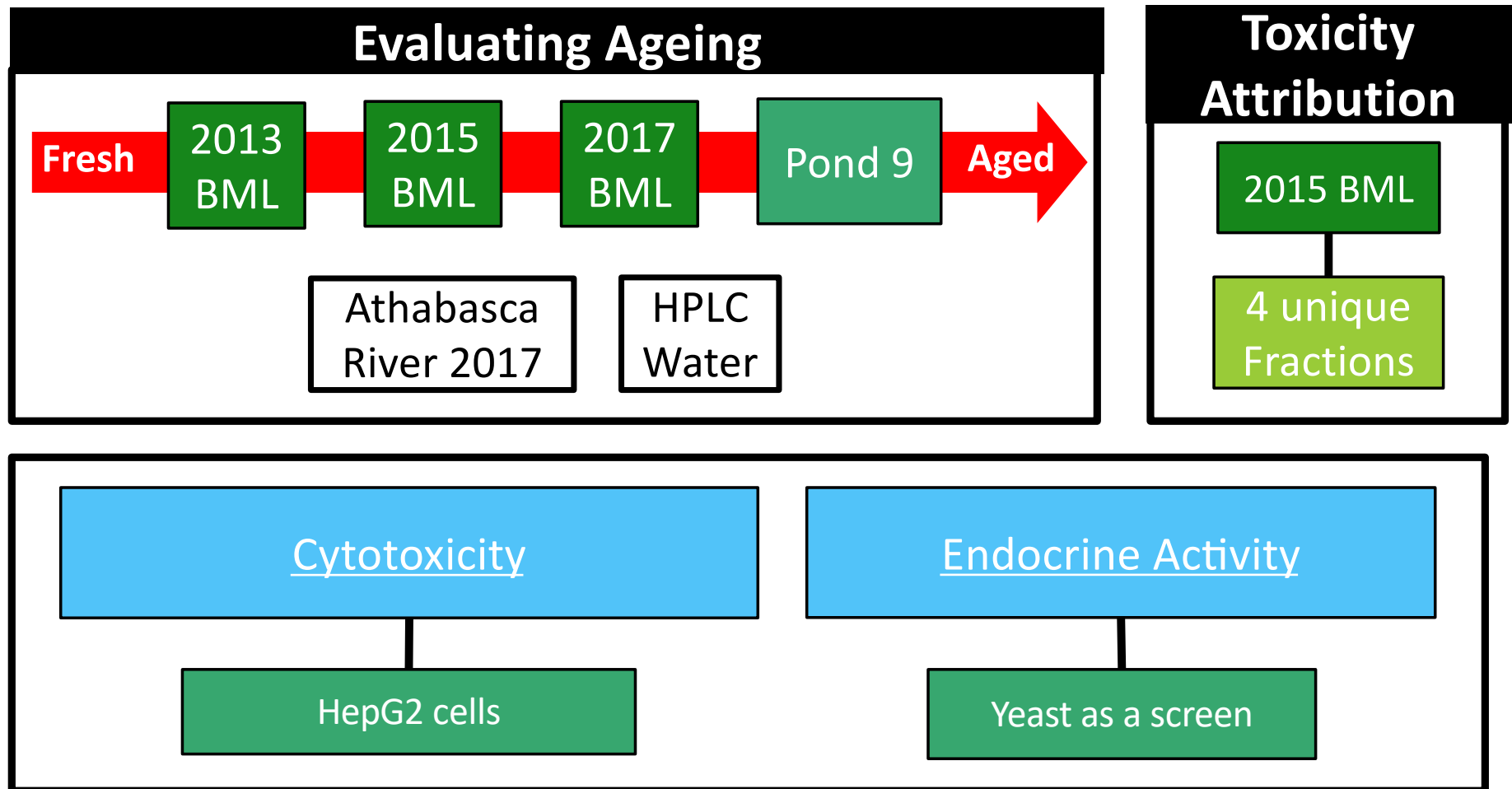


- Need to test human-relevant cytotoxicity, confirm other mode of actions, and look at how toxicity changes over time

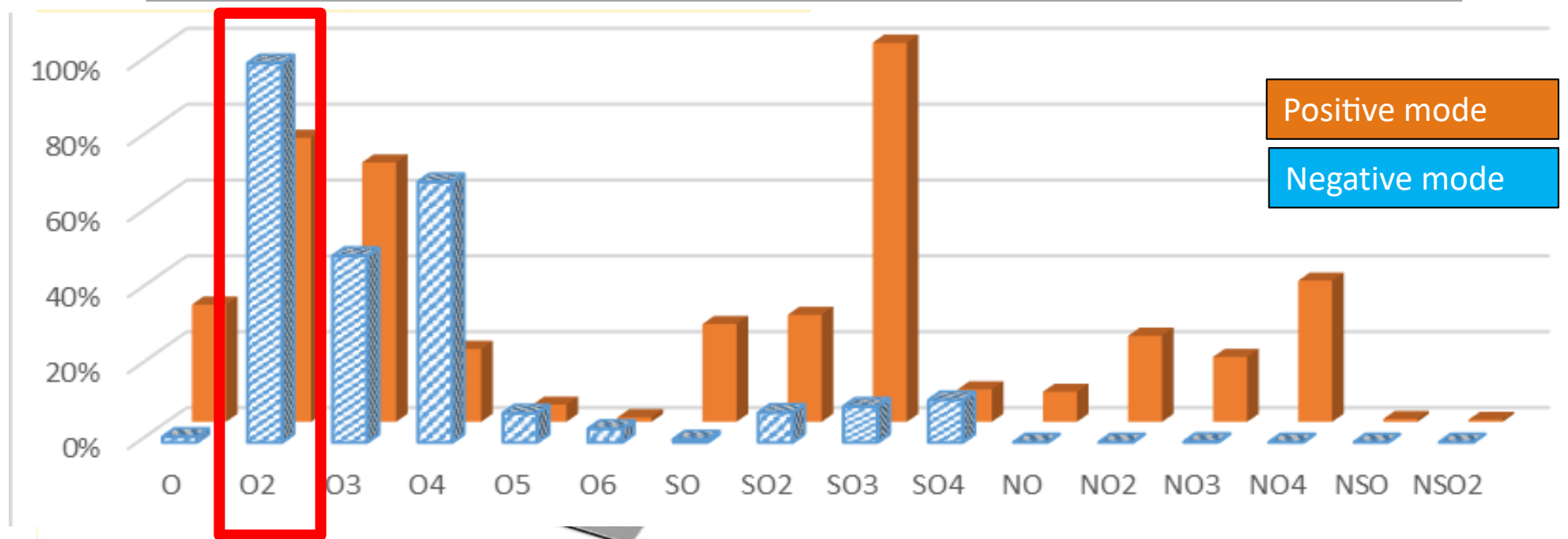
Research Question and Objectives

- In the End Pit Lake strategy, is **ageing** effective in decreasing the toxicity of OSPW organics and what chemical classes need to be targeted to enable a safer release?
- 1. Compare the cytotoxicity and endocrine activity of OSPW at different ages
- 2. To attribute toxicity to particular organic chemical classes

Study Design



Orbitrap Mass Spectrometry: How We Describe the Mixture



Total ion
chromatogram of
bitumen-derived
organics

Assign empirical formulas
based on accurate mass
measurements
($C_xH_yO_zS_aN_b$)

Bin into chemical
classes
(O_x , SO_x , NO_x)

Endocrine Activity:

Yeast Estrogenic & Androgenic Screen (YES/YAS)

- Transactivational assay measuring human Estrogen Receptor (**ER**) & Androgen Receptor (**AR**) interactions
- **OSPW**:
 - Varied results of agonism and antagonism^{1, 2}
- Recurrent **caveats** to warrant further study:
 - Extraction
 - Mass spectrometry
 - Source

Current Status of BML:

Toxicity Data from Syncrude Canada LTD.

Lethal Concentration (LC ₅₀) values* (Units: % BML Water)		
Acute Toxicity Test	2016 Data	Historical Data (2013-2015)
Ceriodaphnia	77%	72%
Daphnia	>100%	>100%
Fathead Minnow	>100%	63%
Rainbow trout	>100%	66%

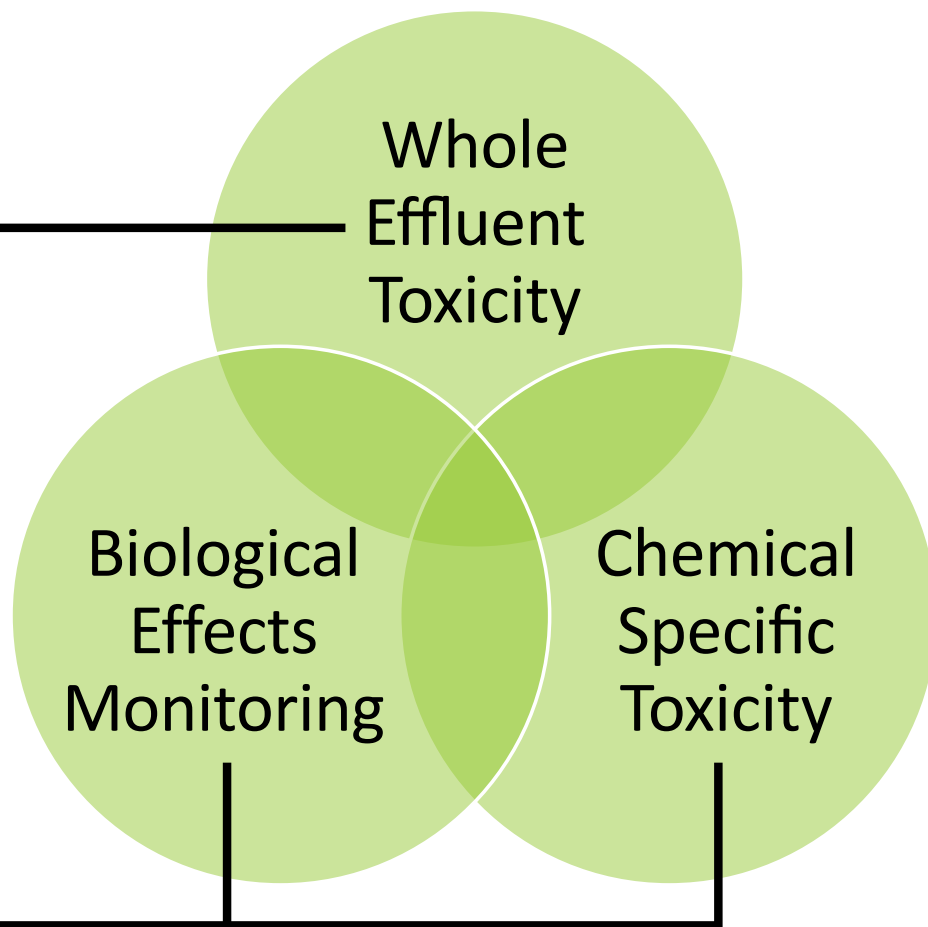
*Minimum, not median LC₅₀ values

Triad Approach to Maximize Environmental Protection

- Water quality-based effluent limits for industrial discharges in Alberta (1995)

Syncrude's toxicity data

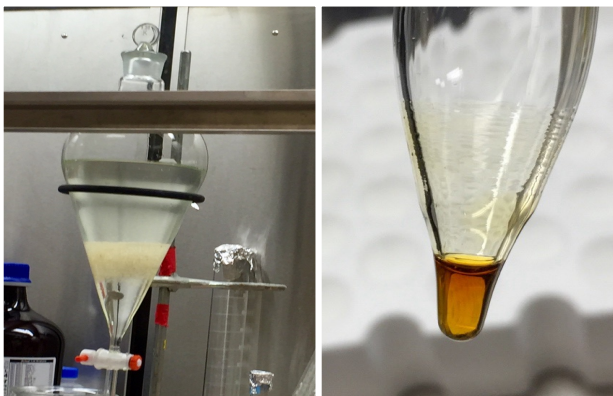
Project: toxicity of whole organic mixture and their fractions; insight into sub-lethal effects



1) Evaluating Ageing as a Remediation Strategy

Sample Preparation:

Assessing the Whole Organic Mixture



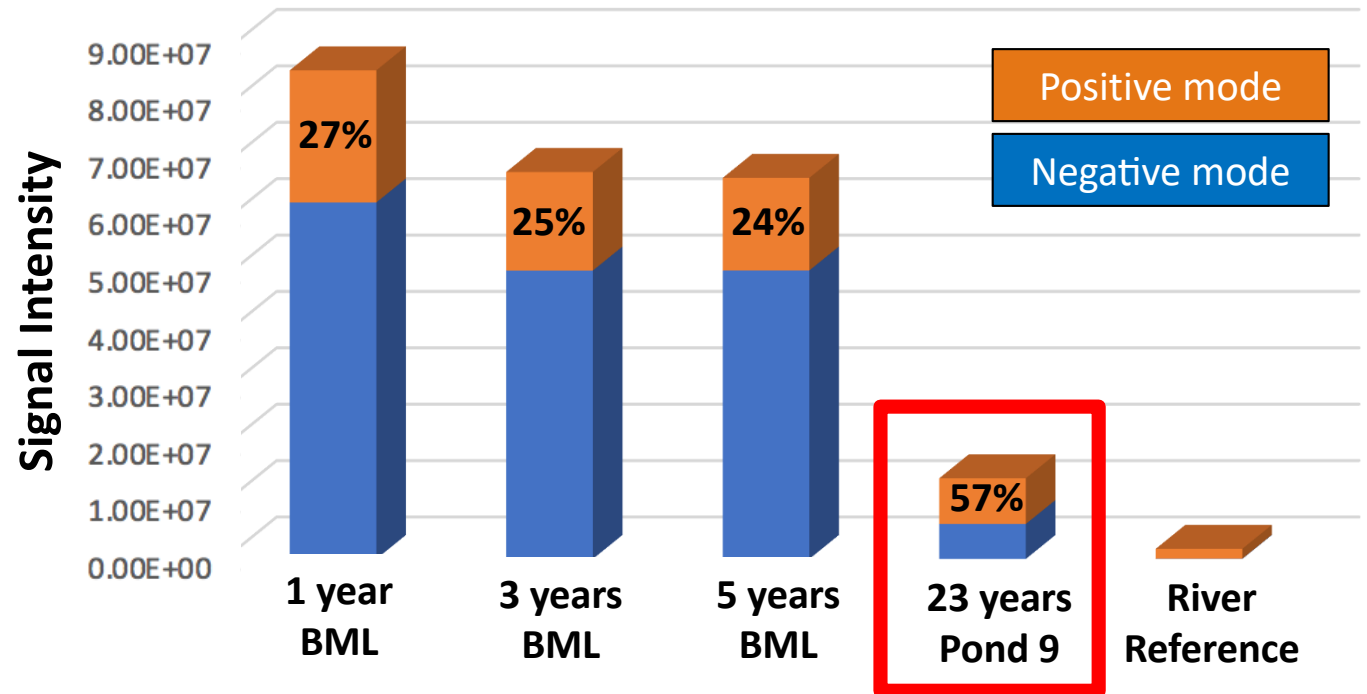
- **1 L** Total Acid Extract (**mg/L**)
- Environmentally relevant **Enrichment Factor**
 - $0.1x \rightarrow 1x \rightarrow 12.5x$
- Concentration vs **composition**

	River	BML 2013	BML 2015	BML 2017	Pond 9
Years Aged		1	3	5	23
Organic [] (mg/L)	0.8	75.8	66.0	57.9	38.1

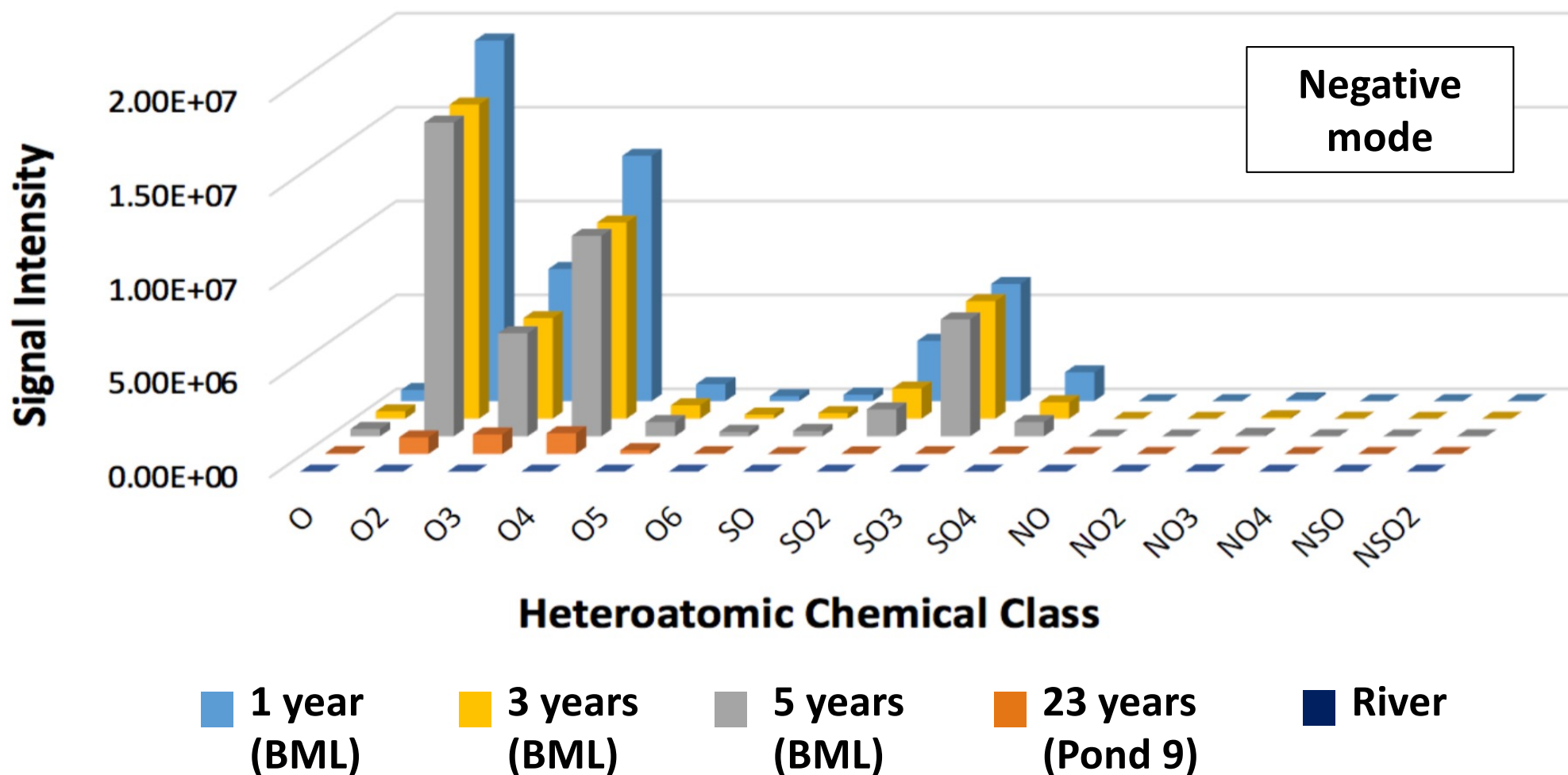
Organic concentration decreases with age

Orbitrap Mass Spectrometry Consistent with Gravimetric Analysis

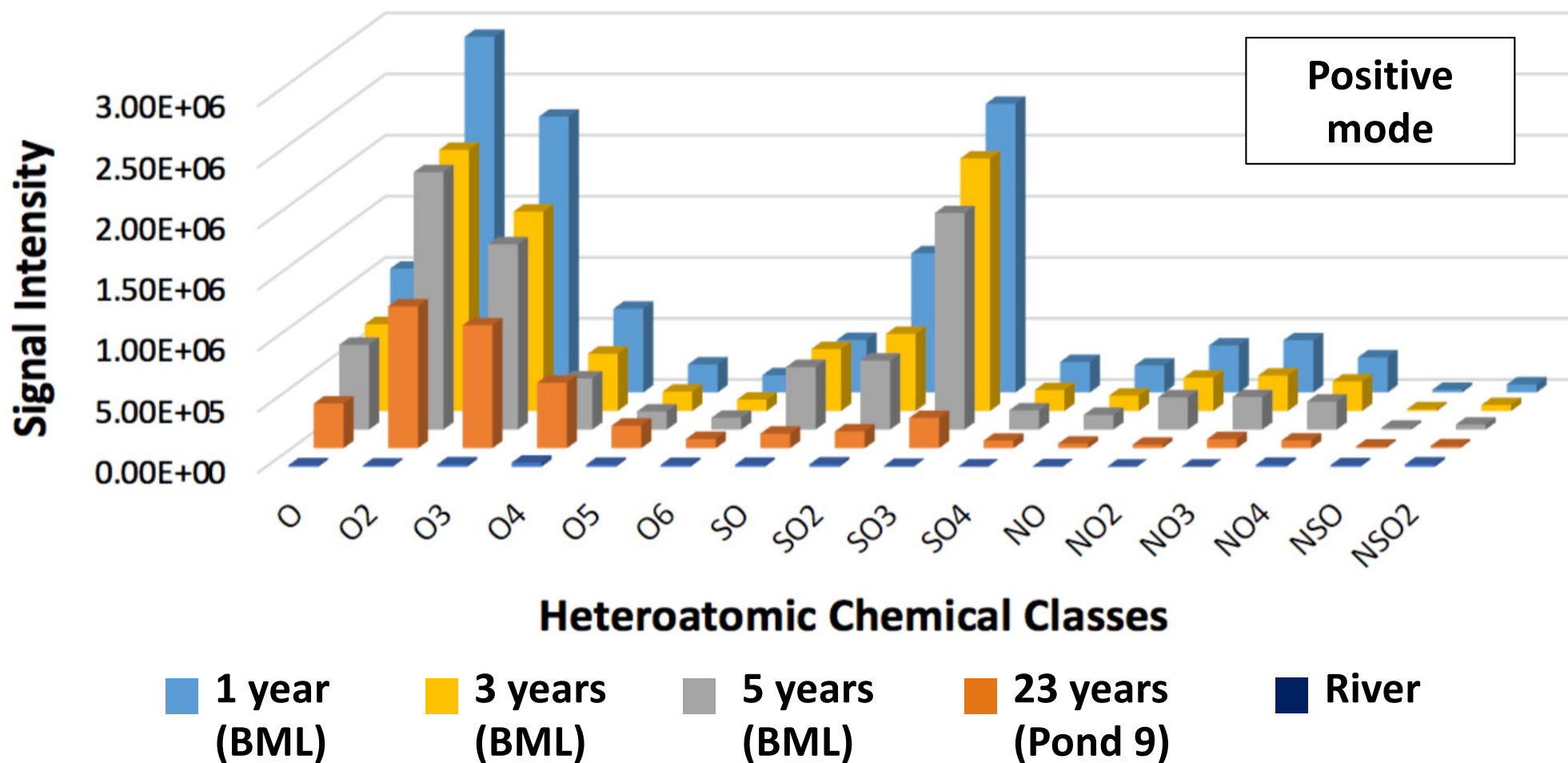
- The organic mass and signal intensity decreases with **ageing**
- **Negative** mode species depleted to greater extent
- **Positive** mode species appear more persistent over time



Beyond Total Intensity: Chemical Class Distributions



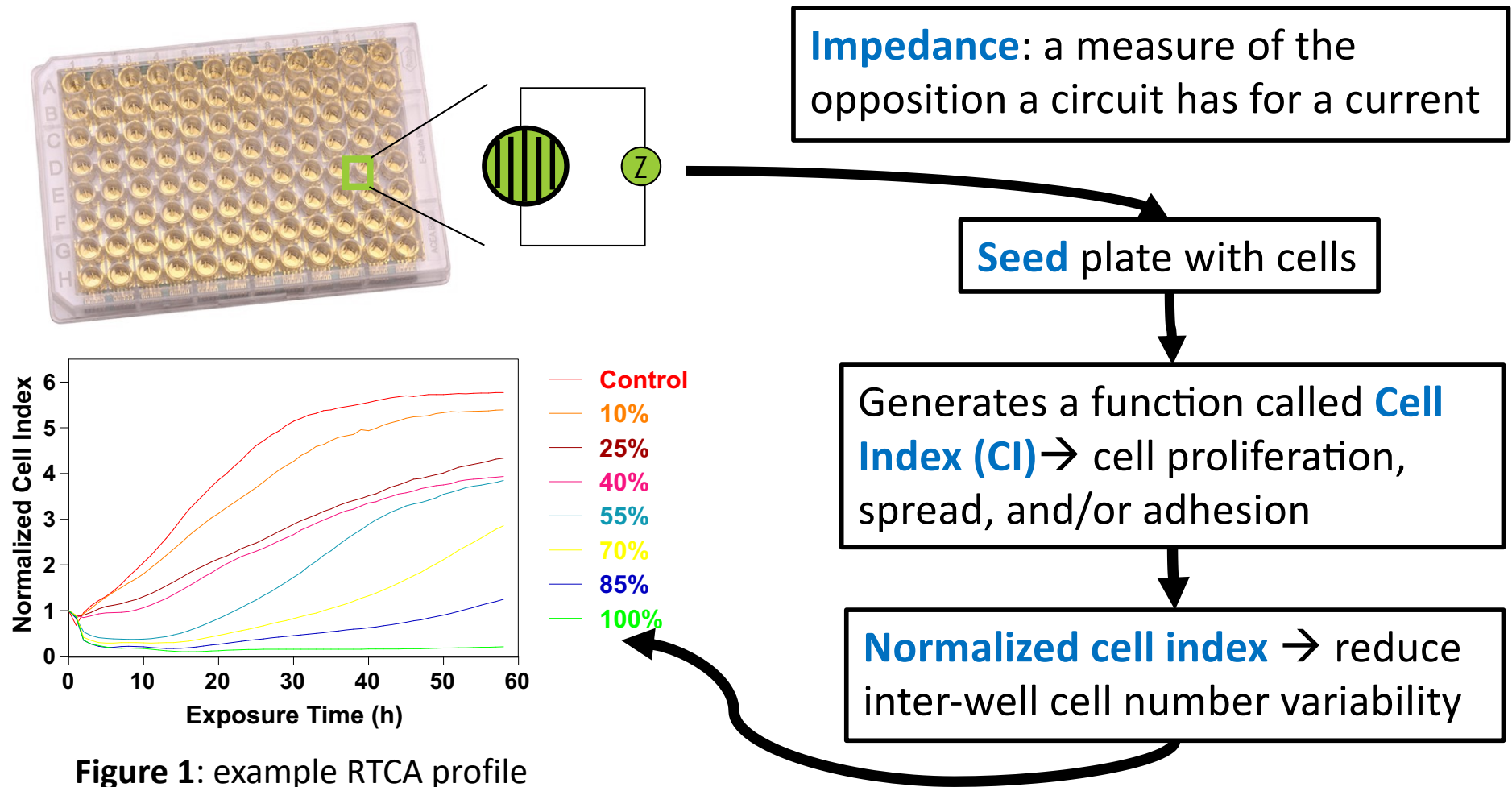
Chemical Class Distributions in Positive Mode



Real-Time Cell Analysis (RTCA)

CONTINUOUS MEASUREMENT OF CELL
PROLIFERATION OVER TIME

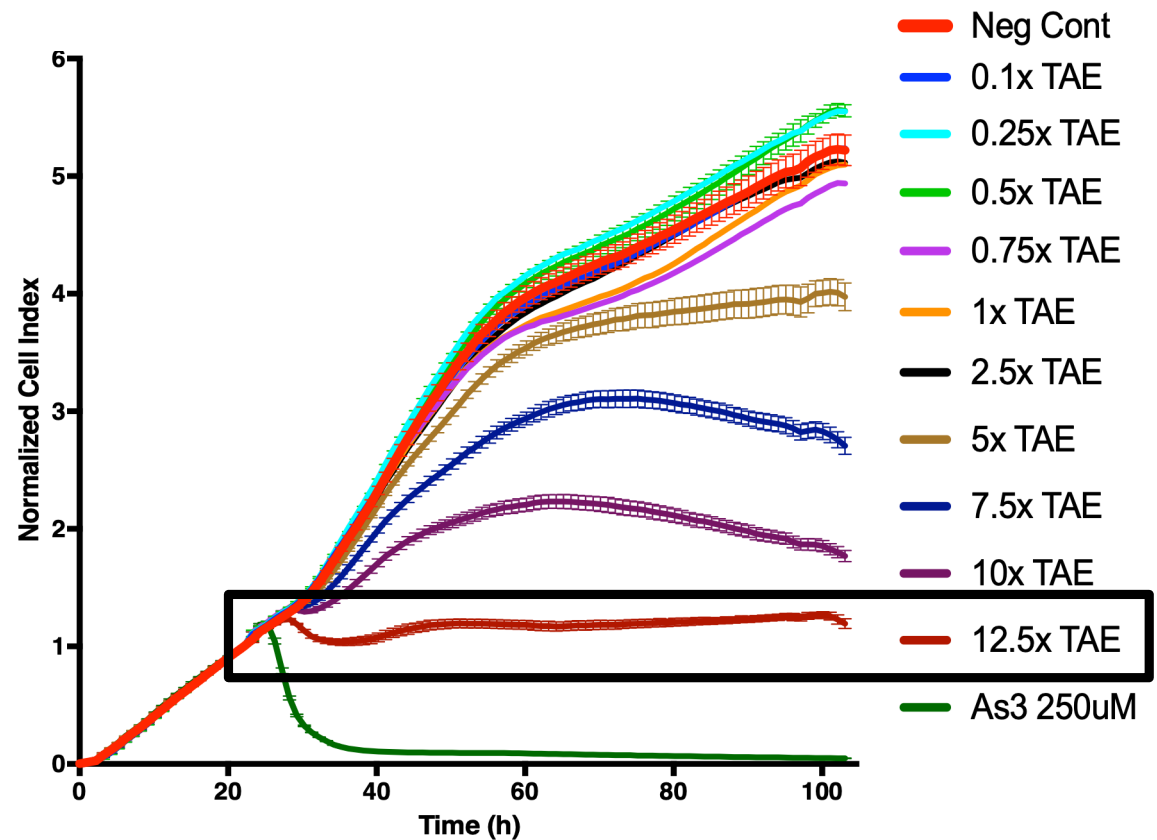
Cytotoxicity through RTCA



BML 2015:

Bitumen-Derived Organics are Cytotoxic

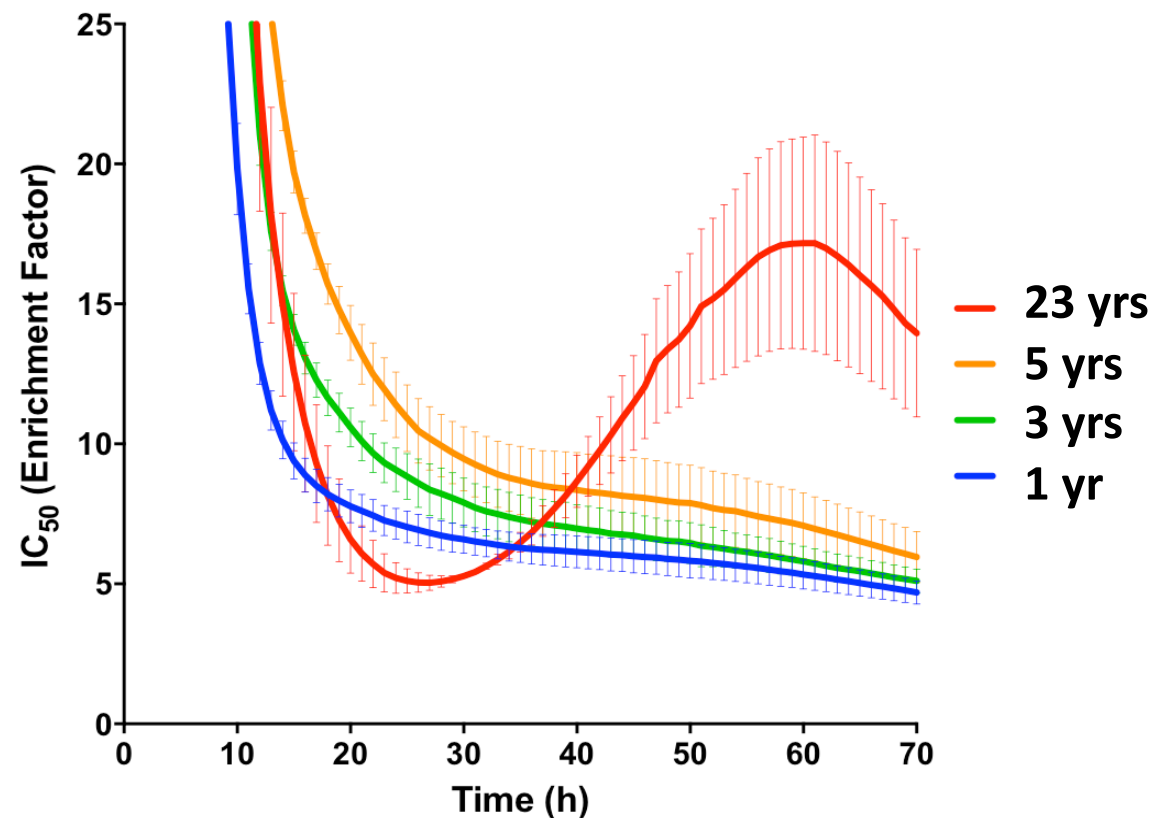
- Extraction control and Athabasca River reference did not affect cell proliferation



IC₅₀ Histogram:

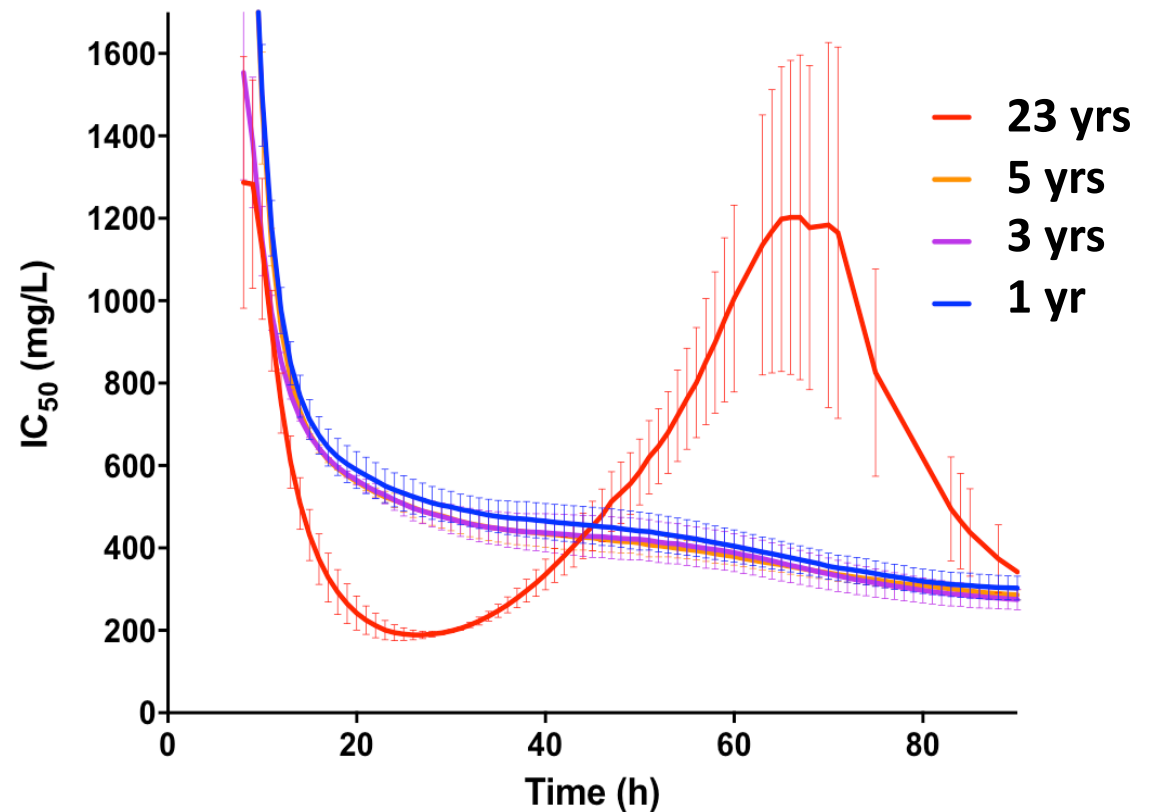
Summary of OSPW of Different Ages

- **Inhibitory Concentration (IC₅₀)**
 - Concentration of an inhibitor that reduces cell growth by half
- **BML** cytotoxicity decreases with age on enrichment factor scale
- **Aged sample:** biphasic toxicity profile



Assessing Toxic Potency: Enrichment Factor $\rightarrow$ mg/L

- Differences in cytotoxicity among BML samples disappear
 - Suggests that **dilution**, not degradation, may be responsible
 - **BML** modeled as receiving fresh water dilution of 5-10% per year^{1,2}
- Recalcitrant organics in **aged sample** are more potent than BML ~24h

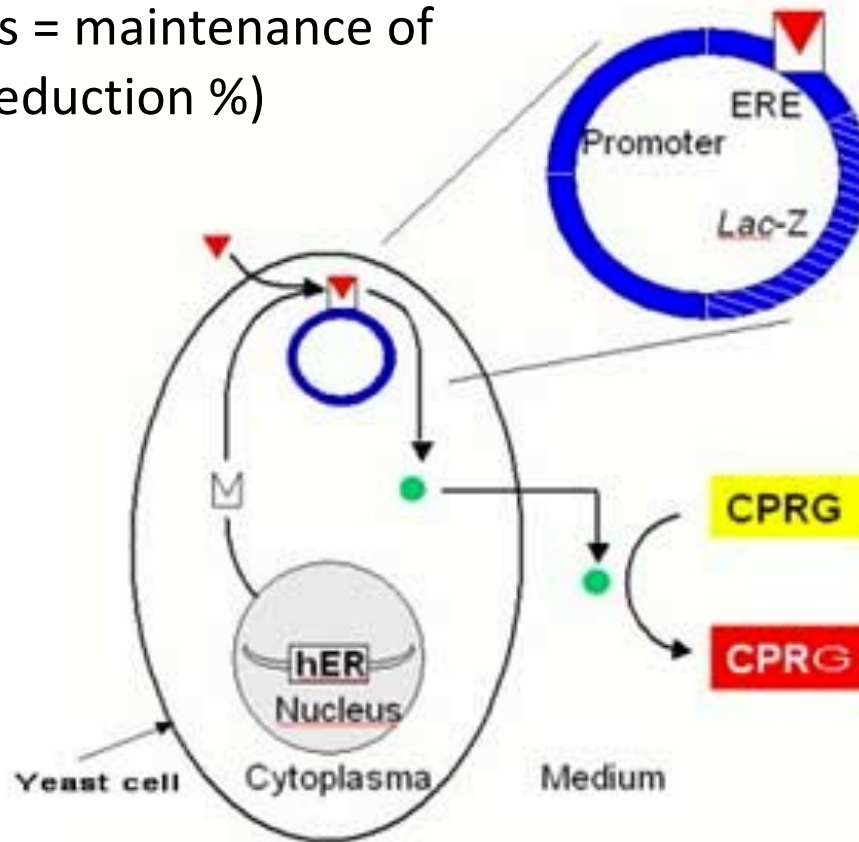


Yeast Estrogenic & Androgenic Screen (YES/YAS)

YES/YAS Method: YES as an Example

Positive Results

- Agonists = **RED** (induction %)
- Antagonists = maintenance of **YELLOW** (reduction %)



▼	Estrogen
□	Estrogen Receptor
▼	Activated Receptor
●	B-Galactosidase

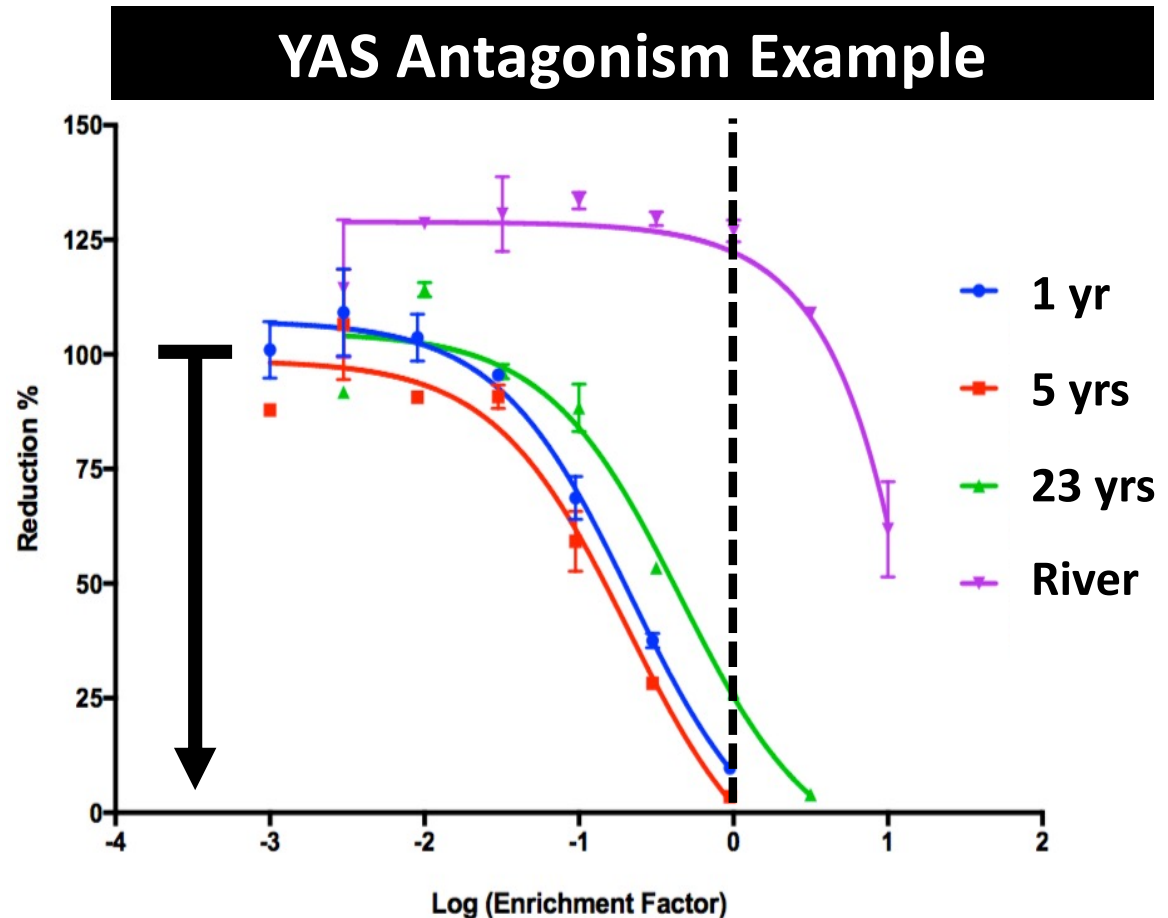
hER	Estrogen Receptor
ERE	Estrogen Response Element
Lac-Z	Reporter Gene
CPRG	Colorimetric indicator

YES/YAS Results: OSPW of Different Ages

- **Sub-lethal doses** (×): 1× BML, 3× Pond 9, 10× River

	YES	YAS
Agonism	×	×
Antagonism	✓	✓

- Divergent from **River**
- Does not change with **age**



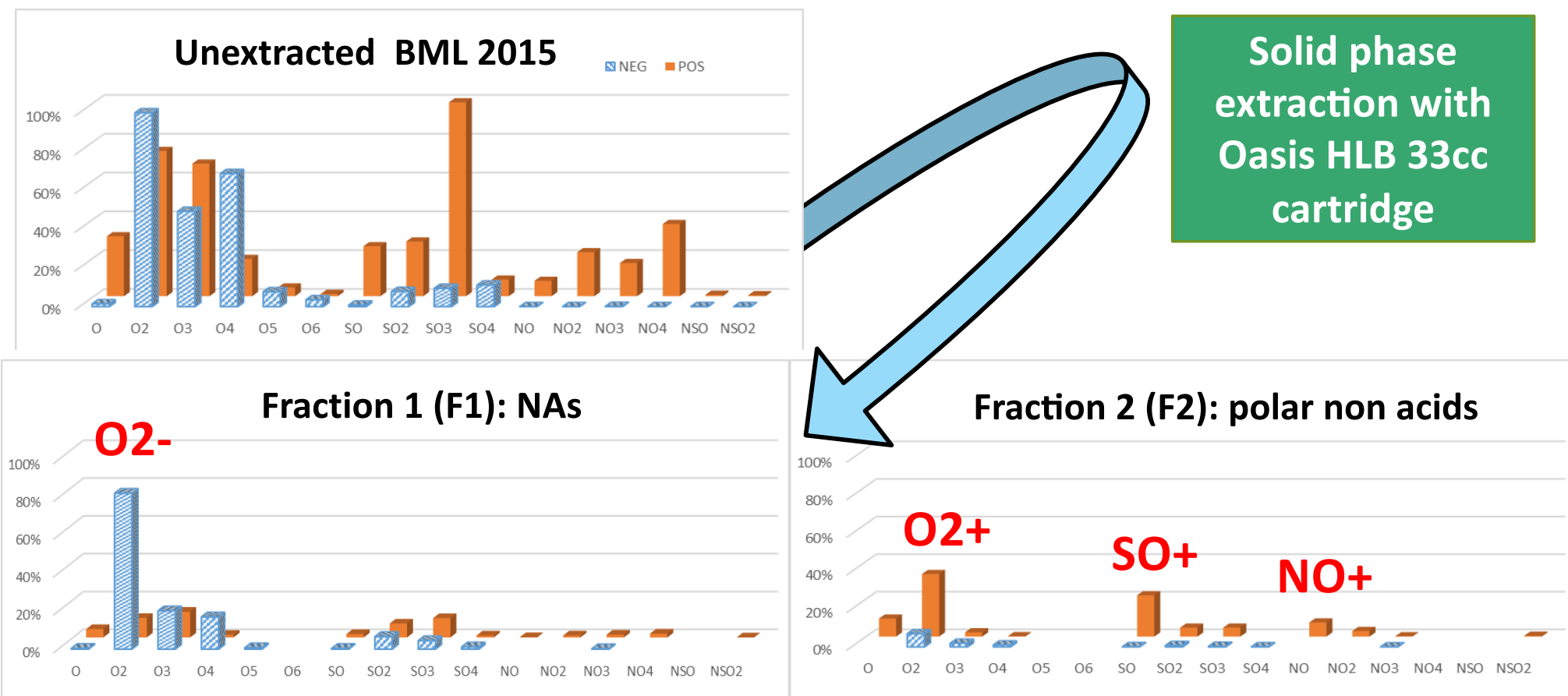
Conclusions:

Evaluating Ageing as Remediation

1. Cytotoxicity of whole BML is decreasing as it ages
 - However, the **toxic potency** of the remaining organics of BML remain unchanged
 - **Dilution**, not degradation may be responsible
2. Most aged sample has biphasic profile; more potent than fresh at 24 hrs
 - **Concerning**, as sample is representation of BML in future
3. OSPW appear endocrine active through **antagonism** of **ER** & **AR**
 - Not ameliorated with **ageing**, despite decrease in mass and signal intensity over time
 - **Positive mode species** more persistent and may maintain the effect

2) Toxicity Attribution of BML 2015 Organic Fractions

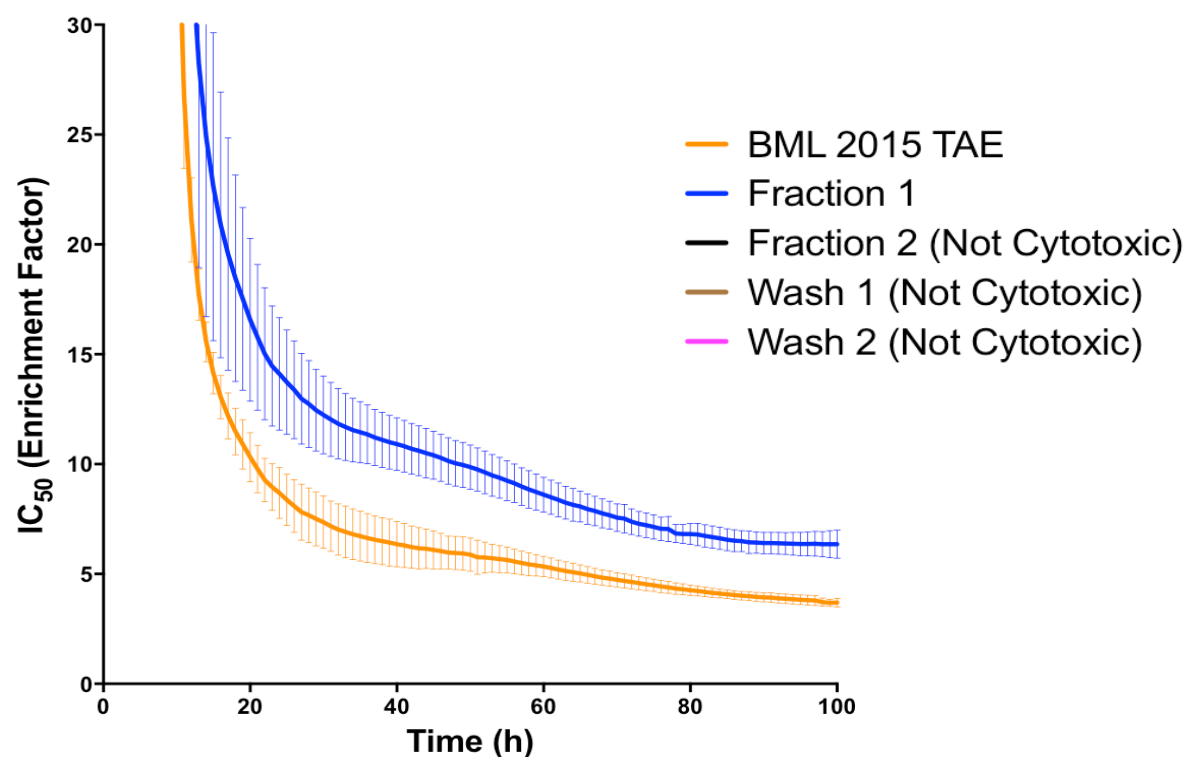
After the Effects-Directed Analysis on BML 2012,¹ have candidate chemical classes guiding fractionation of BML 2015



Two washes containing a mixture of chemicals

Real-Time Cell Analysis (RTCA)

NAs Largely Responsible for BML 2015 Cytotoxicity



- **Fraction 2** from previous study was cytotoxic, not here
 - Alternative mode of action?

Yeast Estrogenic & Androgenic Screen (YES/YAS)

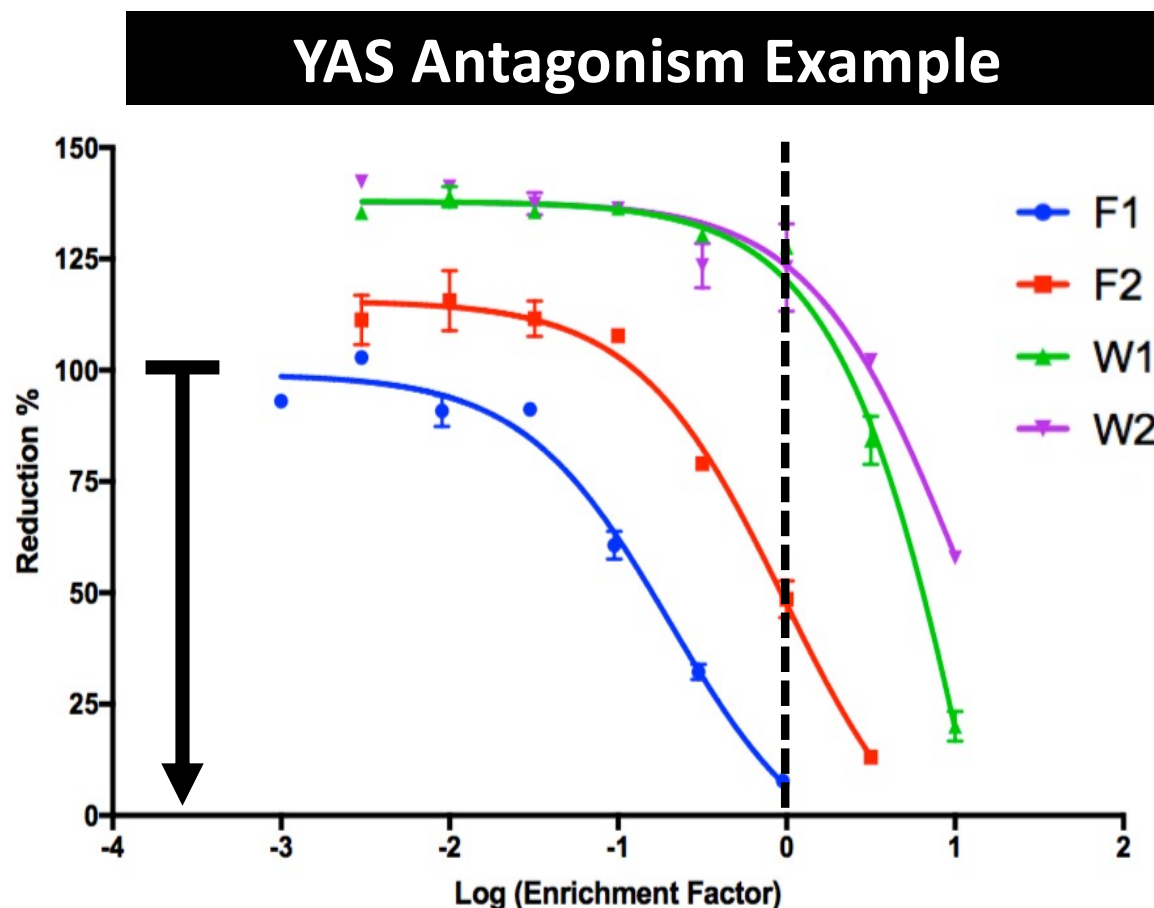
YES/YAS Results:

Toxicity Attribution in BML 2015

- **Sub-lethal doses** (×): 1× F1, 3× F2, 10× W1, W2

	YES	YAS
Agonism	×	×
Antagonism	✓	✓

- Largely isolated endocrine activity



Conclusions:

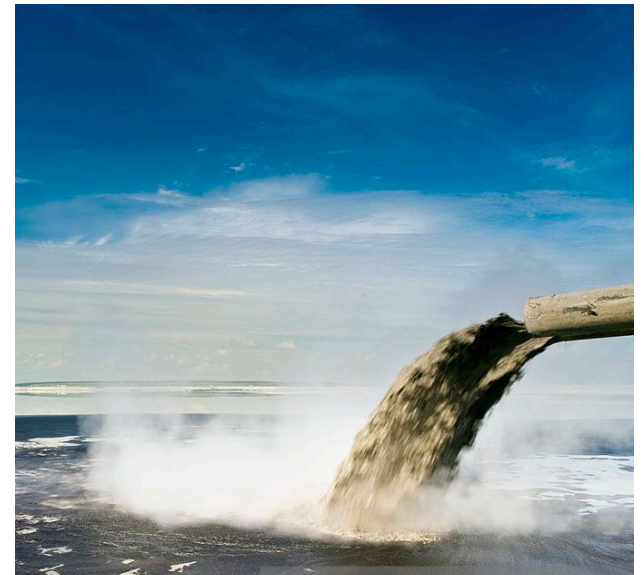
Toxicity Attribution in BML 2015

Divergence in mechanisms between chemical classes

1. **NAs**: most acutely toxic class to a human cell line and yeast
2. All **BML 2015 fractions** are endocrine active through ER & AR antagonism, not agonism
 - **Polar non-acids** and **NAs** most potent; largely isolated endocrine activity
 - Antagonism is **effective** at doses below field concentrations

Environmental Relevance

1. Organic mixture is more persistent than previously thought
2. Polar non-acids may be just as **toxicologically relevant** as NAs due to endocrine activity
3. **Toxic potency**, through both lethal and non-lethal mechanisms, is not being ameliorated with ageing
4. Within tested timeline, **dilution** appears as the only mechanism reducing concentration and toxicity



Is “Dilution the Solution to Pollution?”

Acknowledgements

- Martin Lab
- Li Lab
- Keelan Lab
- Birget Moe
- Angela Sun

